MIL-G-5413D

30 June 1966

Superseding
MIL-G-5413B (ASG)
17 February 1959
and MIL-G-005413C (USAF)
29 October 1963

MILITARY SPECIFICATION

GENERATORS, TACHOMETER, FOUR-POLE

This specification is mandatory for use by
* all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - This specification covers two types of four-pole alternating current tachometer generators.

1.2 Classification - The generator shall be furnished in the following part numbers, as specified:

* MS25038-1	Four-pole tachometer generator which can also operate a propeller synchronizer/synchrophaser.
MS25038-2	Plain four-pole tachometer generator.

* 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited)

FSC 6620

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SPECIFICATIONS (Cont'd)

Federal

QQ-N-290	Nickel Plating (Electrodeposited)
QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-Z-325	Zinc Plating (Electrodeposited)
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000	Drawings, Engineering and Associated List
MIL-M-3171	Magnesium Alloy, Process for Corrosion, Protection of
MIL-C-5015	Connectors, Electric "AN" Type
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-L-6082	Lubricating Oil; Aircraft Reciprocating (Piston) Engine
MIL-S-6872	Soldering Process, General Specification for
MIL-I-7098	Indicator, Tachometer, Electric Magnetic Drag Type, Aircraft General Specification for

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MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-P-23408	Plating: Tin-Cadmium (Electrodeposited)

STANDARDS

Military

AND20005	Drive - Type XV Engine Accessory
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-781	Reliability Tests Exponential Distribution
MS25038	Generator, Tachometer, Four-pole, Propeller Synchronizer
MS33540	Safety Wiring, General Practices for
MS33586	Metals, Definition of Dissimilar

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* (Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 Qualification - The generators furnished under this specification shall be a product which has been tested and meets the Quality Assurance Provisions specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Selection of government documents - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.2.1 and 3.2.2.

3.2.1 Standard parts - With the exception of 3.2.2, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.2.2 Commercial parts - Commercial parts having suitable properties may be used, where on the date of invitation for bids, there are no suitable standard parts. In any case, commercial parts, such as screws, bolts, nuts, cotter pins, having suitable properties shall be used provided:

(a) They can be replaced by the standard parts (MS or AN) without alteration.

(b) The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest weight, and suitable for the purpose intended.

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3.3.1 Critical materials - Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.3.2 Nonmagnetic materials - Nonmagnetic materials shall be used for all parts of the generator except where magnetic materials are essential.

3.3.3 Metals - Metals shall be of the corrosion-resistant type, or shall be suitably protected as specified herein to resist corrosion due to fuels or oils, salt spray or atmospheric conditions to which the generator may be subjected when in storage or during normal service life.

3.3.3.1 Dissimilar metals - Dissimilar metals as defined in MS33586 shall not be used in intimate contact with each other, unless suitably protected against electrolytic corrosion by means of protective coatings.

* 3.3.3.2 Aluminum alloy parts - Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625. Small holes, threads and case inserts need not be anodized. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

* 3.3.3.3 Iron and steel parts - Where practicable, iron and steel parts shall be chromium, nickel, cadmium or zinc plated in accordance with QQ-C-320, QQ-N-290, QQ-P-416 or QQ-Z-325, respectively. Parts in a confined space in the presence of organic material shall be tin-cadmium plated in accordance with MIL-P-23408. The class and type of plating, shall meet the requirements of 3.3.4.

* 3.3.3.4 Magnesium alloy parts - Magnesium alloy parts shall be treated in accordance with MIL-M-3171. When abrasion resistance is a factor, an anodic treatment approved by the procuring activity shall be used. For new design, the use of magnesium is prohibited unless specifically approved by the procuring activity.

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* 3.3.4 Protective treatment - When materials are used in the construction of the generator that are subject to atmospheric or environmental conditions likely to cause corrosion in normal service life, they shall be protected against corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. Finishes and protective coatings which crack, chip or scale during normal service life or are affected by extremes of atmospheric or environmental conditions shall not be used.

3.3.5 Fungus-proof materials - Materials which are nutrients for fungi shall not be used where it is practicable to avoid them. Where used, they shall be treated with a fungicidal agent acceptable to the procuring activity. If used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

* 3.3.6 Fumes and vapors - Materials used in the construction of the generator shall not produce corrosive, deleterious, or toxic fumes or vapors under the conditions specified herein.

* 3.4 Design and construction - The generator shall be designed to operate at any speed up to 2500 RPM in either direction of rotation and in any mounting position. The generator shall conform to MS25038 and shall be suitable for mounting on a pad conforming to AND20005. The generator design shall be such that the output voltage shall be three phase, having a frequency of two cycles per revolution of the generator drive shaft. The generator shall be designed to withstand a steady 20 g load applied along any of the three principal mutually perpendicular axes when mounted on a pad conforming to AND20005. The generator shall be so constructed that no parts will work loose in service and will withstand the normal shocks, vibrations, and such other conditions as are incident to shipping, storage, installation, and service without failure. The generator shall have 1000 hours minimum operating life on an aircraft engine.

3.4.1 Maintenance - The design of the generator shall be such as to accommodate to the greatest possible extent, disassembly, reassembly, and service maintenance using tools and items of maintenance, equipment which are normally available as commercial standards.

3.4.2 Oil seal - The generator shall be provided with an oil seal at the drive end as near the face of the mounting flange as possible. The oil seal shall be capable of preventing engine oil leakage in excess of 2 cubic centimeters per hour when under a pressure of 1 psi.

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3.4.3 Drain holes - The generator shall be provided with four drain holes for the removal of excess engine oil as specified on MS25038.

* 3.4.4 Drive shaft - The drive shaft overall length shall be 1.250 inch (minimum) and shall fit inside the main shaft. The drive shaft shall be secured by a pin 0.078 inch diameter (minimum) and the pin located so that it is held in place by the drive end bearing.

3.4.4.1 Shaft hardness - The hardness of the drive end of the shaft shall be Rockwell C-30 to C-40.

* 3.4.4.2 Shaft strength - The drive shaft shall have a shear section of uniform cross-sectional area for a minimum of 0.25 inch of shaft length. The proportional elastic limit (P.E.L.) shall not be exceeded until a torque greater than 30 inch-pounds is applied. The shaft shall then withstand a twist greater than 270 degrees prior to failure. The shaft shall fail within the torque limits of 35 to 45 inch-pounds.

* 3.4.4.3 Backlash - The shaft rotational backlash about its longitudinal axes shall not exceed 6 angular degrees.

* 3.4.5 Stator lock - The stator assembly shall have a locking means to prevent rotation between the stator and case or end housing. A press fit shall not be considered a locking means.

* 3.4.6 Rotor - The magnet may be pressed on to the rotor shaft, however, a press fit shall not be relied on as the sole means of preventing relative movement between the magnet and shaft.

* 3.4.7 Bearings - The bearings used in the generator shall be standard commercial bearings obtainable directly from the bearing manufacturer.

* 3.4.7.1 Mounting - Non-separable bearings mounted in aluminum or other non-ferrous material require steel bearing liners. Separable bearings may use the outer ring in lieu of a liner provided an interference fit is used.

* 3.4.7.2 Location (MS25038-1) - The outer ring of the bearing shall be located in the end housings and the bearing seats shall be integral with the end housings. The stator assembly shall serve as a means of alignment for the end housings and bearings.

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3.5 Weight - The weight of the generator shall not exceed 2.1 pounds for the MS25038-2 generator and 4.5 pounds for the MS25038-1 generator.

3.6 Screw threads - Screw threads 0.06 inch in diameter or larger, shall be in accordance with MIL-S-7742. All screws used in the construction of the generator shall be safety wired in accordance with MS33540.

3.7 Electrical requirements -

3.7.1 Electric connectors - The generator shall be provided with an electric connector conforming to MIL-C-5015, except that the insert shall be of a resilient polychloroprene base compound and the connector shall be suitably protected against corrosion. The size and type of connector shall be as specified on MS25038.

3.7.2 Electrical connections - The three leads from the generator windings shall be connected to B, A, and C pins of the electric connector in such a manner that when the generator drive shaft is rotated in a counterclockwise direction (looking into the spline end of the shaft) the generator output phase sequence will be B, A, C. The lead connected to pin C of the connector shall be grounded to the housing of the generator.

* 3.7.3 Wire - All electrical wire, including both winding and connecting wire, shall be made of copper.

* 3.8 Performance - The generator shall perform satisfactorily when subjected to the inspections specified in Section 4 of this specification.

* 3.8.1 Reliability -

* 3.8.1.1 Operational stability - The generator shall operate with satisfactory performance, continuously or intermittently for a period of at least 1000 hours without the necessity for any adjustment.

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* 3.8.1.2 Reliability in Mean Time Between Failure (MTBF) - The generator shall have 1000 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

* 3.9 Identification of product -

* 3.9.1 Nameplate - A nameplate shall be securely attached to the exterior of the case and shall be marked in accordance with the requirements of MIL-STD-130 except that the FSN shall be omitted.

* 3.9.2 Manufacturer's part number - The manufacturer's part number on the nameplate shall be identical with the manufacturer's engineering production drawing number including applicable dash numbers if the drawing is tabulated and covers more than one part.

* 3.10 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-1000.

3.11 Soldering - Soldering shall be performed in accordance with MIL-S-6872.

* 3.12 Workmanship - The generator including all parts and accessories shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine-screw assemblage, welding and brazing, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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* 4.2 Classification of inspection - Inspection of the generator shall be classified as follows:

(a) Qualification inspection: Qualification inspection consists of examinations and tests performed on sample generators submitted for approval as a qualified product.

(b) Quality conformance inspection: Quality conformance inspection consists of examinations and tests performed on generators manufactured and submitted for acceptance under contract.

* 4.3 Qualification inspection - The qualification inspection of the generator shall consist of all of the examinations and tests of this specification performed in the order specified under the paragraph headed Inspection methods and shall include the reliability test for the Reliability Qualification Phase. Unless otherwise stated in the contract the tests of the Reliability Sampling Phase only are required.

* 4.3.1 Qualification inspection sample - Qualification inspection samples shall consist of two generators manufactured in accordance with this specification. The generators submitted for qualification inspections shall have been previously subjected only to the Individual inspections. The samples shall be forwarded at the contractor's expense, to the laboratory designated in the Letter of Authorization.

* 4.3.1.1 Qualification inspection sample identification - The qualification inspection samples shall be plainly identified by durable tags, securely attached, and marked with the following information:

Sample for Qualification Inspection

Generator, Tachometer, Four-Pole

MS25038 - (Add proper dash no.)

Submitted by (Manufacturer's *name, date*)
for Qualification Inspection in accordance
with MIL-G-5413 under authorization (reference
letter authorizing tests).

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* 4.4 Quality conformance inspection - The Quality conformance inspection shall consist of the Individual Inspection, the Sampling plans, and the Reliability assurance tests of this specification. The contractor shall furnish all samples and shall be responsible for accomplishing all the inspections. Sampling plan B inspections shall be conducted at a Government laboratory designated by the procuring activity or as directed by the procuring activity. Quality conformance inspection, except for Sampling plan B, when performed by a Government laboratory, shall be under the supervision of the Government quality control representative. The contractor shall furnish test reports showing quantitative results for all tests required by this specification, signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product.

* 4.4.1 Individual inspection - Each generator submitted for acceptance shall be subjected to the Individual inspection. This inspection shall determine compliance with the requirements of material, workmanship, operational adequancy, and reliability. As a minimum, each generator accepted shall have passed the following inspections:

Examination of product.

Continuity and resistance.

Phase rotation.

Voltage.

* 4.4.2 Sampling plans - The sampling plans shall consist of Sampling plan A and Sampling plan B inspections. The inspection samples selected for sampling tests shall first have passed the Individual inspections. The inspection samples which have been subjected to Sampling plan A inspection shall not be delivered on contract until they have been refurbished (see 6.3) resubmitted and passed all the Individual inspections. Inspection samples which have been subjected to the Sampling plan B inspection shall not be delivered on contract.

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* 4.4.2.1 Sampling plan A sampling selection - Sampling plan A samples shall be selected at random in accordance with the following schedule:

Quantity of Generators Offered for Acceptance	Quantity to be Selected for Testing
First 15	SEE NOTE
Next 50	1
Next 75	1
Next 100	1
Each additional 200 or fraction thereof	1

Note: When Sampling plan B is invoked, the quantity shall be zero. When Sampling plan B is to be omitted, the quantity shall be one.

When a defective generator occurs, no items from those still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. In addition, when a failure occurs, shift to one sample out of fifteen (when Sampling plan B is omitted) and proceed as indicated.

4.4.2.1.1 Sampling plan A inspection - Each sample selected for Sampling plan A inspection shall be subjected to the following tests in the order listed.

Vibration endurance

Dielectric strength

Oil leakage

* 4.4.2.2 Sampling plan B instructions - Two generators shall be selected at random from the first 15 produced on contract. These samples shall be submitted within 10 days after manufacture, at the contractor's expense, to a Government laboratory designated by

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the procuring activity or as otherwise directed by the procuring activity. Each sample shall be plainly identified by a durable tag, securely attached and marked with the following information:

(Generator, Tachometer, Four-Pole MS25038 - (add proper dash No.)

Submitted by Manufacturer's name, date) for Production Acceptance Sampling Plan B inspection, in accordance with Contract/order No.

Manufacturer's part number _____

* 4.4.2.2.1 Sampling plan B approval - Approval of sampling plan B generators shall be by the procuring activity upon satisfactory completion of the designated tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the sample units to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished.

* 4.4.2.2.2 Sampling plan B inspection - Each generator selected for Sampling plan B inspection shall be subjected to the following tests, conducted in the order listed:

Sampling plan A Inspection

Low Temperature

High Temperature

Acceleration

Short circuit

Overspeed and reverse operation

Harmonic content (MS25038-1 only)

Endurance

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Oil leakage retest

Internal examination

Shaft failure

* 4.4.3 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Qualification Phase and Sampling Phase Tests shall be conducted. Equipments selected for Reliability Assurance Tests shall first have passed the Individual tests.

* 4.4.3.1 Reliability Qualification Phase - Prior to Qualified Products Listing (see 4.3), a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests." The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E (temperature only) shall be used. The Accept-Rejection Criteria for Test Plan I shall be used.

* 4.4.3.2 Reliability Sampling Phase Tests - Samples of the equipment shall be tested as outlined in MIL-STD-781, under the Section entitled "Sampling Phase of Production Reliability Tests." For the Sampling Phase, Test Level E (temperature only) shall be used. The Accept-Reject Criteria for Test Plan II shall be used to determine the length of the tests (until an accept or reject decision is reached).

* 4.4.3.2.1 Lot size for sampling phase - The entire contract quantity (with the exception of those equipments used in the Qualification Phase) shall be considered one lot. The test program shall start the first month after the Qualification Phase has been completed. Samples shall be selected and placed on test in the same manner as though one month's production was a lot. The samples shall be tested until an accept or rejection decision is reached. In the meantime other equipments constructed shall be shipped. The test results shall be summarized monthly for the procuring activity. The procuring activity reserves the right to stop the acceptance of equipment at any time after one or more reject decisions have been reached, pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment quality control etc, so that the entire contract quantity will show an accept decision.

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* 4.4.3.3 Test details - The test details such as the length of the Test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test etc. shall be part of the test procedures to be submitted to and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests. The following paragraphs shall be considered minimum conditions and shall apply to both the Qualification and Sampling phases.

* 4.4.3.3.1 Duty cycle - The duty cycle shall be as conducted in 4.6.14.

* 4.4.3.3.2 Performance characteristics to be measured - The requirements of 4.6.14 apply.

* 4.4.3.3.3 Failure criteria - Whenever performance characteristics fall below the acceptance requirements (4.4.3.3.2) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each should be counted a failure, unless one part caused the other parts to fail.

4.4.3.4 Preventive maintenance - No preventive maintenance shall be performed on the generators during the Reliability test.

* 4.4.3.3.5 Disposition of samples upon completion of tests - Any sample tested other than those tested under the Reliability Qualification Phases may be delivered on contract provided it meets all the following requirements:

(a) It is representative of production units currently being accepted.

(b) It is in "good as new" condition or has been refurbished. (See 6.3).

(c) It is otherwise satisfactory.

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4.5 Inspection conditions -

4.5.1 Standard conditions - Unless otherwise specified, all tests required shall be made under the following conditions:

Temperature - Room ambient (25±5°C).

Pressure - Normal atmospheric (approximately 29.92 inches Hg).

Humidity - Room ambient up to 90 percent relative humidity

* 4.5.2 Voltages - All voltages specified herein are RMS values. Voltmeters used to measure voltage shall be of a type free from frequency errors. If the voltmeter draws appreciable current, the specified loads shall be so adjusted that the combined voltmeter and loads will form a balanced load equivalent to the specified load. The voltmeter shall have an accuracy of 1% or better and meter errors shall not be included in the errors attributable to the generator. Unless otherwise specified, all voltage readings shall be made after a five, and before ten, minute generator warm-up with Standard load at Standard speed.

4.5.3 Test indicator - Whenever a tachometer indicator is specified for conducting tests, a magnetic drag-type electric tachometer indicator in accordance with MIL-I-7098 shall be used.

4.5.4 Standard load - Whenever a standard load is specified herein, it shall consist of three 40-ohm wye-connected resistors for the MS25038-2 generator and three 30-ohm delta-connected resistors for the MS25038-1 generator.

4.5.5 Standard speed - The standard generator speed as specified herein shall be 1250 RPM.

4.6 Inspection methods -

4.6.1 Examination of product - Each generator shall be examined externally to determine conformance to the applicable drawings and to all the requirements of the specification not covered by tests.

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* 4.6.2 Continuity and resistance - The windings of the generator shall be checked for continuity between the A, B, and C pins of the electric connector, and between each of the A, B, and C pins and the generator case. There shall be no evidence of discontinuity. The resistance between each pair of pins of the MS25038-2 generator shall be within 2.0 ohm of the others. The resistance between each pair of pins of the MS25038-1 generator shall be within 0.20 ohm of the others.

* 4.6.3 Phase rotation - The generator shall be connected to an indicator conforming to MIL-I-7098. The generator terminals A, B, and C, shall be connected to the indicator terminals A, B, and C, respectively. The indicator shall show a positive reading when the generator drive shaft is rotated in a counterclockwise direction as viewed looking at the spline end of the generator shaft.

4.6.4 Voltage -

4.6.4.1 MS25038-2 generator - With the generator connected to a standard load and operating at the standard speed, no one of the three terminal voltages of the generator shall exceed 21 volts when measured at the generator terminals. With the generator connected to a load of three 20-ohm wye-connected resistors and operating at a shaft speed of 300 rpm, no one of the three terminal voltages shall be less than 3.5 volts.

4.6.4.2 MS25038-1 generator - With the generator connected to a standard load and operating at the standard speed, each of the three terminal voltages of the generator shall be between 19.5 and 21 volts when measured at the generator terminals. With the generator connected to a load of three 15-ohm delta-connected resistors and operating at a shaft speed of 300 rpm, no one of the three terminal voltages shall be less than 3.5 volts.

* 4.6.5 Vibration endurance - The generator shall be subjected to linear vibration in any direction perpendicular to the horizontal axis as shown on MS25038. The frequency shall be cycled through a range of from 10 to 200 to 10 cps at the rate of 1 cycle in 15 minutes. Between 10 and 30 cps, the double amplitude shall be held constant at

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0.25 inch and between 30 and 200 cps, the acceleration shall be held constant at 20 g ± 2g. The duration of the test shall be 24 hours which may be conducted during 3 periods of 8 hours each. The generator shall be mounted on the vibration machine and connected to operate two indicators which conform to MIL-I-7098. During vibration, the generator shall be operated continuously, its shaft speed being cycled through a range of 300 to 1500 to 300 RPM at the rate of 1 cycle in 10 minutes. During the test the indicators shall be observed for any erratic operation which would indicate a generator malfunction. After the completion of 24 hours of vibration, the generator shall be allowed to cool to room temperature. The generator shall then be subjected to, and meet, the requirements of 4.6.2 and 4.6.4. There shall be no mechanical failure or loosening of parts as a result of vibration.

4.6.6 Dielectric strength - A potention of 500 volts at a commercial frequency shall be applied between any pin of the electric connector and any metal part of the generator case or the electric connector shell for 5 seconds. The C lead shall be disconnected from the housing of the generator for this test. There shall be no breakdown of insulation.

4.6.7 Oil leakage - The generator shall be operated at 1200 ± 50 RPM with the shaft in a horizontal position and with one of the drain holes located at the bottom. While operating at this speed, the drive shaft end of the generator shall be subjected to an oil pressure of 1 psi using oil conforming to MIL-L-6082, at a temperature of 120° ± 5°C. The amount of oil leakage out of the drain hole shall be determined at the end of 1 hour after the first drop of oil appears at the drain hole. This oil leakage shall not exceed 2 cubic centimeters for the 1-hour period. If no oil appears at the drain hole after 1 hour of operation under the above conditions, the test shall be considered complete.

CAUTION: Provide adequate ventilation during this test.

4.6.8 Low temperature - The generator shall be subjected to the Low Temperature Tests, Procedure II, of MIL-E-5272. At the end of the 48-hour period, while the generator is still at the low temperature, the torque required to rotate the generator shaft shall be measured and shall not exceed 12 inch-pounds. At the end of the 24-hour

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period, the torque shall again be measured and shall not exceed 8 inch-pounds. After the chamber has been returned to room temperature and the generator has saturated at room temperature for a minimum period of 4 hours, it shall be subjected to and shall meet the requirements of the tests listed in 4.4.1. There shall be no damage which would adversely affect subsequent operation.

* 4.6.9 High temperature - During the high temperature test, the generator shall be connected to operate two indicators conforming to MIL-I-7098 at a generator speed of 1200 ± 50 rpm and the output voltages shall be noted. The generator shall then be subjected to a temperature of 90° ± 2°C for 4 hours and while at this temperature the generator shall operate the same two indicators at the same speed. Near the end of the test the output voltages shall be measured. The average of the voltages measured at the high temperature shall not differ from the average of the voltages measured at room temperature by more than 2 volts. No damage to any part of the generator shall result from this test. After a minimum of three hours at room temperature the generator shall be subjected to, and meet, the requirements of 4.6.4.

4.6.10 Acceleration - The generator drive shaft shall be coupled to a motor which starts with an initial acceleration of 400 revolutions per second per second, and the generator shall be subjected to this acceleration for 500 applications. The final speed of the motor on each run shall be 1,000 to 1,800 rpm. No damage to any part of the generator shall result from this test.

4.6.11 Short circuit - The generator shall be connected to the standard load and operated at the standard speed. The generator shall be short-circuited five times for a period of 1 second each time across all three of the generator terminals with an interval of 5 seconds between short circuits. After the generator has been allowed to cool to room temperature, the generator shall be subjected to, and meet, the requirements of 4.6.2 and 4.6.4.

4.6.12 Overspeed and reverse operation - The generator shall be connected to the standard load for the standard speed. The generator shall then be operated with the same load at 2,500 rpm for a period of 5 minutes, and then in the reverse direction at 1.250 rpm for a period of 1 minute. After the generator has been allowed to cool to room temperature, the generator shall be subjected to, and meet, the requirements of 4.6.4.

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* 4.6.13 Harmonic content - The MS25038-1 generator shall be connected to a standard load and operated at the standard speed. The harmonic content shall be determined and shall not exceed 10 percent total distortion as determined by the following formula or equivalent:

$$\text{Percent Distortion (Total)} = \frac{(\text{sum of squares of amplitudes of harmonics})^{1/2}}{\text{Amplitude of fundamental}} \times 100$$

* 4.6.14 Endurance - The generator shaft rotational backlash shall be measured and shall be within the requirements of 3.4.4.3. The generator shall then be mounted on an especially constructed simulated engine accessory drive generally in accordance with AND 20005. The drive shall have its centerline displaced 0.007 inch from the centerline of the pilot and the generator mounting pad face shall be canted 0.008 inch TIR out of square with the centerline. The generator shall then be connected to a load consisting of three delta connected 30-ohm - 100-watt resistors. The generator shall be operated on this drive and with this load for 1000 hours at the following shaft speeds and continuous cycle.

350 ± 50 RPM one minute
1200 ± 50 RPM four minutes

At the conclusion of each 100 hours, the generator shall be subjected to, and meet, the requirements of 4.6.4 and of 3.4.4.3.

No screw or other part shall become loosened or damaged as a result of this test.

4.6.15 Humidity - This test shall be in accordance with the Humidity Tests, Procedure I, of MIL-E-5272, except that the cycle shall be repeated a sufficient number of times to extend the total time of test to 120 hours (5 cycles). At the end of the 120-hour (5 cycle) period, within 1 hours after the generator has been removed from the chamber, the generator shall be subjected to and shall meet the requirements of the test specified in 4.6.6. Examination for corrosion and other deterioration shall be made between 24 and 48 hours after completion of the humidity exposure. There shall be no evidence of corrosion or other deterioration. The generator shall then be subjected to, and meet, the requirements of 4.6.4.

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4.6.16 Salt spray - The generator, with external connections made to simulate installed conditions, shall be subjected to the Salt Spray Tests in accordance with Procedure I of MIL-E-5272. There shall be no evidence of corrosion or other deterioration. At the end of the test, the generator shall be subjected to and shall meet the requirements of the tests specified in 4.6.2, 4.6.4, and 4.6.6.

4.6.17 Oil leakage retest - The generator shall be subjected to, and meet the requirements of 4.6.7 except that oil leakage shall not exceed 5 cubic centimeters in one hour..

4.6.18 Internal examination - Examination of the disassembled generator shall show no evidence of excessive wear or other deterioration.

* 4.6.19 Disassembly - reassembly - The generator shall be disassembled to the extent that all fastened parts, except press or shrink fitted parts, are separated. The fastening means shall be noted and all shall be in accordance with Requirement 12 of MIL-STD-454, except that Sealing Compounds shall not be used. The generator shall then be completely reassembled reusing all fasteners of the original assembly except that new safety wire shall be used. The generator shall then be operated to determine that it runs properly and that all parts are satisfactorily mated.

The disassembly and reassembly as described above shall be done five times on each generator and the same requirements shall be met.

* 4.6.20 Shaft failure - The generator shall be disassembled and the shaft of the armature caused to fail. Failure shall occur within the torque limits of 35 to 45 inch-pounds.

5. PREPARATION FOR DELIVERY

* 5.1 Packaging - The generator shall be packaged in accordance with MIL-P-7936, Level A or C as specified in contract or order (see 6.2). For Level A packaging the method of preservation shall be in accordance with MIL-P-116, Method 1A-5, without preservative compound using metal reuseable containers, or Method 1A8 without preservation compound, overboxed in a container conforming to PPP-B-636, whichever is specified in contract or order.

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5.2 Packing - The generator shall be packed in accordance with MIL-P-7936, Level A, B or C as specified in contract or order (see 6.2).

* 5.3 Marking - The interior and exterior containers shall be marked as specified in MIL-P-7936.

* 5.3.1 Precautionary marking - The following precautionary markings shall appear conspicuously on two opposite sides of each interior package and shipping container wherever practicable.

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended use - MS25038-1 generators covered by this specification are intended for use on reciprocating engines to generate electrical signals for transmission to indicators conforming to MIL-I-7098 and also to propeller synchronizers. MS25038-2 generators covered by this specification are intended for use on reciprocating engines to generate electrical signals for transmission to indicators conforming to MIL-I-7098.

* 6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) The quantity, MS part number, dash number, and type number, if any, of the instrument desired.

(c) Levels of packaging (see 5.1) and packing (see 5.2) desired.

(d) The laboratory that shall conduct tests (see 4.3.1, 4.4, 4.4.2.2). Unless specifically waived, Sampling Plan B Tests will be conducted.

(e) Reliability Test Phases (see 4.3).

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* 6.3 Definitions -

* 6.3.1 Amplitude - Whenever the word "amplitude" is used it shall mean the extent of motion as measured from one extreme to the opposite extreme.

* 6.3.2 Good as new - "Good as new" shall mean generators operated with less than 10 percent of the specified MTBF operation.

* 6.3.3 Refurbished - "Refurbished" shall mean that the generator has been completely overhauled with all component parts meeting current part standards; and the generator shall have been subjected to and met all the requirements of a new generator.

* 6.4 Changes from previous issue - The outside margins of this specification have been marked "*" to indicate where changes from the MIL-G-5413B issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

* 6.5 Precedence of documents - When the requiremnts of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specification.

(b) This specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(c) Referenced specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therin. All referenced specifications shall apply to the extent specified.

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6.6 Qualification - With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for the opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they proposed to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Air Systems Command, Department of the Navy, Washington, D. C 20360.

* Custodians:
Air Force - 67
Army - MO
Navy - AS

* Preparing Activity
AS

* Review activities:
Air Force - 67
Army - MO
Navy - AS

* Project No. 6620-0094

* User Activities:
Air Force -
Army -
Navy -

* Reviewer/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the Current Federal Supply Classification Listing of DOD Standardization Documents.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-G-5413D Generators, Tachometer, Four-Pole

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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Naval Air Engineering Center
Philadelphia, Pa. 19112

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Naval Air Engineering Center

Philadelphia, Pennsylvania 19112

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